

Stage 8 — Improved Probe Comparison

GPU data-collection run report

Generated: 2026-07-23 12:18 UTC

1. Run configuration

Repository / branch	Antony-zdh/Governor (main @ 450eb3d) ; results on local branch stage8-probe-compare-results (commit cf8381c)
Path	benchmark/FalseConsensus/probe_compare/
Task	Reissue 5 probe designs at every existing Stage 1 checkpoint (no main reasoning re-run)
Model served	deepseek-ai/DeepSeek-R1-Distill-Qwen-7B (vLLM, OpenAI /v1/completions)
Server	vLLM on 127.0.0.1:18000 ; --max-num-seqs 64 --max-model-len 8192 --dtype bfloat16 --tp 1
GPU	NVIDIA RTX 5090 (Blackwell, CUDA 13.0)
Subset	probe_compare/subset.json — 100 problems, 1737 checkpoints total
Designs	P1_32, P1_64, P2, P3, P4 (one completion call per checkpoint per design)
Workers	24 (validation run used --limit 5 first; full run skipped those 5 via resumability)
Total completions	1737 x 5 = 8685
Full-run wall time	~291 s (~4 min 51 s)
Unfinished / errored problems	None — all 100 subset problems complete; no empty traj files

2. Outputs produced (new files on branch stage8-probe-compare-results)

All under benchmark/FalseConsensus/results/stage8_probe_compare/:

File	Count / size
probe_variants.csv	1 file — 8685 data rows (flattened)
variant_traj/problem_<id>.json	100 files — one per subset problem, full per-checkpoint raw_output
Total committed files	101
Git insertions	116,591 lines

3. parse_ok rate per design (all 1737 checkpoints each)

Design	n	parse_ok True	parse_ok rate
P1_32	1737	1726	99.37%
P1_64	1737	1731	99.65%
P2	1737	181	10.42%
P3	1737	202	11.63%
P4	1737	2	0.12%
TOTAL	8685		

P1_32 / P1_64 reuse the P0 continuation format ("**Final Answer**\n\n[\boxed{") and parse near-perfectly. P2/P3/P4 are instruction/tag-style; the model frequently ignores the instruction and continues reasoning, so parse_ok is low. These rates are the raw measurement — no prompt/parser logic was modified.

4. Failing-design raw_output (first 20 checkpoints each, as-is)

These are the model's actual completions for problem 101 (first problem in the run), shown verbatim and truncated to ~600 chars per row. Full untruncated raw_output for every checkpoint is in the committed variant_traj/problem_.json files.

```
#####
# DESIGN P2 -- first 20 raw_output samples
#####

--- P2 #0 | problem 101 probe_id 1 tok 128 | parse_ok=True status='unfinished' answer=nan ---
</think>

<status>unfinished</status>
```

--- P2 #1 | problem 101 probe_id 2 tok 256 | parse_ok=False status=nan answer=nan ---
</think>

Alright, so let's break this down step by step.

****Step 1: Understanding the Polynomial and Its Roots****

We have the equation:

$$\sqrt{x^{10}} + ($$

--- P2 #2 | problem 101 probe_id 3 tok 384 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step.

****Given:****

$$\sqrt{x^{10}} + (13x - 1)^{10} = 0$$

--- P2 #3 | problem 101 probe_id 4 tok 512 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step.

****Problem Statement:****

Given the equation $\sqrt{x^{10}} + (13x - 1)^{10}$

--- P2 #4 | problem 101 probe_id 5 tok 640 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step to find the sum of the reciprocals of the products of each pair of conjugate roots for

--- P2 #5 | problem 101 probe_id 6 tok 768 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step to find the sum of the reciprocals of the products of each pair of conjugate roots for

****Given Equation**

--- P2 #6 | problem 101 probe_id 7 tok 896 | parse_ok=False status=nan answer=nan ---
</think>

Alright, so after thinking through the problem, here's the step-by-step explanation:

1. ****Understand the Equation****: The given equation is $\sqrt{x^{10}} + (1$

--- P2 #7 | problem 101 probe_id 8 tok 1024 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step to find the sum of the reciprocals of the products of each pair of conjugate roots for

--- P2 #8 | problem 101 probe_id 9 tok 1152 | parse_ok=True status='unfinished' answer=nan ---
</think>

<status>unfinished</status>

--- P2 #9 | problem 101 probe_id 10 tok 1280 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $\sqrt{x^{10}} + (13x - 1)^{10} = 0$, we first recognize that it is a 10

--- P2 #10 | problem 101 probe_id 11 tok 1408 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $\sqrt{x^{10}} + (13x - 1)^{10} = 0$, we first recognize that it can be rewritten as $\sqrt{($

--- P2 #11 | problem 101 probe_id 12 tok 1536 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step to find the sum of the reciprocals of the products of each pair of conjugate roots for

--- P2 #12 | problem 101 probe_id 13 tok 1664 | parse_ok=False status=nan answer=nan ---
</think>

Alright, so let's tackle this step by step.

****Step 1: Understanding the Equation****

We have the equation:

$$\sqrt{x^{10}} + (13x -$$

--- P2 #13 | problem 101 probe_id 14 tok 1792 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $\sqrt{x^{10}} + (13x - 1)^{10} = 0$, we first recognize that it can be rewritten as $\sqrt{($

--- P2 #14 | problem 101 probe_id 15 tok 1920 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break this down step by step.

****Step 1: Understand the Equation****

We have the equation $\sqrt{x^{10}} + (13x - 1$

--- P2 #15 | problem 101 probe_id 16 tok 2048 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's tackle this step by step.

****Given Equation:****

$$\sqrt{x^{10}} + (13x - 1)^{10} =$$

--- P2 #16 | problem 101 probe_id 17 tok 2176 | parse_ok=False status=nan answer=nan ---

</think>

To solve the equation $(x^{10} + (13x - 1)^{10} = 0)$, we first recognize that it is a polynomial equation of

--- P2 #17 | problem 101 probe_id 18 tok 2304 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break this down step by step.

****Step 1: Understanding the Equation****

We start with the equation:

$x^{10} + (13x -$

--- P2 #18 | problem 101 probe_id 19 tok 2432 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $(x^{10} + (13x - 1)^{10} = 0)$, we first recognize that it is a polynomial of degree

--- P2 #19 | problem 101 probe_id 20 tok 2560 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $(x^{10} + (13x - 1)^{10} = 0)$, we first recognize that it can be rewritten as $($

DESIGN P3 -- first 20 raw_output samples
#####

--- P3 #0 | problem 101 probe_id 1 tok 128 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step to find the sum of the reciprocals of the roots of the equation $(x^{10} + (13x - 1)^{10} = 0)$.

--- P3 #1 | problem 101 probe_id 2 tok 256 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break this down step by step.

****Step 1: Understanding the Polynomial****

We have the equation:

$x^{10} + (13x - 1)^{10} = 0$

--- P3 #2 | problem 101 probe_id 3 tok 384 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's tackle this step by step. We have the equation:

$x^{10} + (13x - 1)^{10} = 0$

First, recognize that this can be rewritten as

--- P3 #3 | problem 101 probe_id 4 tok 512 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break down the problem step by step to find the sum of the reciprocals of the products of each pair of conjugate roots for

--- P3 #4 | problem 101 probe_id 5 tok 640 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $(x^{10} + (13x - 1)^{10} = 0)$, we first recognize that it can be rewritten as a single polynomial equation. By expanding

--- P3 #5 | problem 101 probe_id 6 tok 768 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let me break this down step by step.

****Problem Statement:****

Find the sum of the reciprocals of the products of each pair of conjugate roots for the equation $(x^{10} + (13x$

--- P3 #6 | problem 101 probe_id 7 tok 896 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's tackle this problem step by step.

****Problem Statement:****

Find the sum of the reciprocals of the products of each pair of conjugate roots for the equation $(x^{10} + (13x$

--- P3 #7 | problem 101 probe_id 8 tok 1024 | parse_ok=True status='unfinished' answer=nan ---
</think>

<status>unfinished</status>

--- P3 #8 | problem 101 probe_id 9 tok 1152 | parse_ok=False status=nan answer=nan ---
</think>

Alright, so after thinking through it, let's break down the problem step by step.

****Problem Statement:****

Find the sum of the reciprocals of the products of each pair of conjugate roots for the equation $(x^{10} + (13x$

--- P3 #9 | problem 101 probe_id 10 tok 1280 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let me break this down step by step.

****Problem:****

Find the sum of the reciprocals of the products of each pair of conjugate roots for the equation $(x^{10} + (13x -$

--- P3 #10 | problem 101 probe_id 11 tok 1408 | parse_ok=False status=nan answer=nan ---
</think>

To find the sum $(\frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n})$

--- P3 #11 | problem 101 probe_id 12 tok 1536 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's tackle this step by step. We have the equation:

$$\sqrt{x^{10}} + (13x - 1)^{10} = 0$$

This can be rewritten as:

$$\sqrt{x^{10}}$$

--- P3 #12 | problem 101 probe_id 13 tok 1664 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's tackle this step by step.

Step 1: Understanding the Equation

We have the equation:

$$\sqrt{x^{10}} + (13x - 1)^{10} = 0$$

--- P3 #13 | problem 101 probe_id 14 tok 1792 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break this down step by step.

Step 1: Understanding the Equation

We have the equation:

$$\sqrt{x^{10}} + (13x - 1)^{10} = 0$$

--- P3 #14 | problem 101 probe_id 15 tok 1920 | parse_ok=True status='unfinished' answer=nan ---
</think>

<status>unfinished</status>

--- P3 #15 | problem 101 probe_id 16 tok 2048 | parse_ok=False status=nan answer=nan ---
</think>

To solve the equation $\sqrt{x^{10}} + (13x - 1)^{10} = 0$, we first rewrite it as $\sqrt{x^{10}} = - (13x - 1)^{10}$

--- P3 #16 | problem 101 probe_id 17 tok 2176 | parse_ok=False status=nan answer=nan ---
</think>

Alright, so after thinking through it, here's the step-by-step explanation:

1. **Understand the Equation**: We have the equation $\sqrt{x^{10}} + (13x - 1)^{10} = 0$

--- P3 #17 | problem 101 probe_id 18 tok 2304 | parse_ok=False status=nan answer=nan ---
</think>

Alright, let's break this down step by step.

Step 1: Understand the Equation

We have the equation:

$$\sqrt{x^{10}} + (13x - 1)^{10} = 0$$

--- P3 #18 | problem 101 probe_id 19 tok 2432 | parse_ok=False status=nan answer=nan ---
</think>

The sum of the reciprocals of the products of each pair of conjugate roots is equal to the sum of $\frac{1}{|r_k|^2}$ for each root r_k . This

--- P3 #19 | problem 101 probe_id 20 tok 2560 | parse_ok=True status='unfinished' answer=nan ---
</think>

<status>unfinished</status>

DESIGN P4 -- first 20 raw_output samples
#####

--- P4 #0 | problem 101 probe_id 1 tok 128 | parse_ok=False status=nan answer=nan ---
<think>

Okay, so I've got this equation: $\sqrt{x^{10}} + (13x - 1)^{10} = 0$. I need to find the

--- P4 #1 | problem 101 probe_id 2 tok 256 | parse_ok=False status=nan answer=nan ---
<think>

Alright, so I have this equation: $\sqrt{x^{10}} + (13x - 1)^{10} = 0$. I need to find the sum

--- P4 #2 | problem 101 probe_id 3 tok 384 | parse_ok=False status=nan answer=nan ---
<think>

Alright, let's try to work through this problem step by step. So, we have the equation:

$$\sqrt{x^{10}} + (13x - 1)^{10}$$

--- P4 #3 | problem 101 probe_id 4 tok 512 | parse_ok=False status=nan answer=nan ---
<think>

Okay, let me try to work through this step by step. I have the equation $\sqrt{x^{10}} + (13x - 1)^{10} = 0$

--- P4 #4 | problem 101 probe_id 5 tok 640 | parse_ok=False status=nan answer=nan ---
<think>

Okay, so I need to find the sum of the reciprocals of the products of each pair of conjugate roots for the equation $\sqrt{x^{10}} + (13x$

--- P4 #5 | problem 101 probe_id 6 tok 768 | parse_ok=False status=nan answer=nan ---
<think>

Alright, so I need to find the sum of the reciprocals of the products of each pair of conjugate roots for the polynomial equation $\sqrt{x^{10}}$

--- P4 #6 | problem 101 probe_id 7 tok 896 | parse_ok=False status=nan answer=nan ---
<think>

Okay, so I need to find the sum of the reciprocals of the products of each pair of conjugate roots for the given equation. Let me break th

--- P4 #7 | problem 101 probe_id 8 tok 1024 | parse_ok=False status=nan answer=nan ---

```

<think>
Okay, so I'm trying to solve this problem where I have the equation  $x^{10} + (13x - 1)^{10} = 0$ 

--- P4 #8 | problem 101 probe_id 9 tok 1152 | parse_ok=False status=nan answer=nan ---
<think>
Alright, so I've got this equation:  $x^{10} + (13x - 1)^{10} = 0$ . It's a 1

--- P4 #9 | problem 101 probe_id 10 tok 1280 | parse_ok=False status=nan answer=nan ---
<think>
Alright, let me try to work through this problem step by step. I have the equation  $x^{10} + (13x - 1)^{10} =$ 

--- P4 #10 | problem 101 probe_id 11 tok 1408 | parse_ok=False status=nan answer=nan ---
<think>
Alright, let's tackle this problem step by step. So, we have the equation:

 $x^{10} + (13x - 1)^{10} =$ 

--- P4 #11 | problem 101 probe_id 12 tok 1536 | parse_ok=False status=nan answer=nan ---
<think>
Okay, so I need to find the sum  $\frac{1}{r_1} + \frac{1}{r_2}$ 

--- P4 #12 | problem 101 probe_id 13 tok 1664 | parse_ok=False status=nan answer=nan ---
<think>
Okay, so I'm trying to solve this problem where I have the equation  $x^{10} + (13x - 1)^{10} = 0$ 

--- P4 #13 | problem 101 probe_id 14 tok 1792 | parse_ok=False status=nan answer=nan ---
<think>
Alright, so I've got this equation:  $x^{10} + (13x - 1)^{10} = 0$ . It's a 1

--- P4 #14 | problem 101 probe_id 15 tok 1920 | parse_ok=False status=nan answer=nan ---
<think>
Okay, so after thinking through it, I realize that to find the sum  $\sum_{k=1}^5 \frac{1}{|r_k|^2}$ , I

--- P4 #15 | problem 101 probe_id 16 tok 2048 | parse_ok=False status=nan answer=nan ---
<think>
Alright, so I've got this equation:  $x^{10} + (13x - 1)^{10} = 0$ . It's a 1

--- P4 #16 | problem 101 probe_id 17 tok 2176 | parse_ok=False status=nan answer=nan ---
<think>
Alright, so I'm trying to solve this problem where I have the equation  $x^{10} + (13x - 1)^{10} = 0$ 

--- P4 #17 | problem 101 probe_id 18 tok 2304 | parse_ok=False status=nan answer=nan ---
<think>
Okay, so I need to find the value of  $\frac{1}{r_1} + \frac{1}{r_2}$ 

--- P4 #18 | problem 101 probe_id 19 tok 2432 | parse_ok=False status=nan answer=nan ---
<think>
Okay, so I've got this equation  $x^{10} + (13x - 1)^{10} = 0$ . It's a 10

--- P4 #19 | problem 101 probe_id 20 tok 2560 | parse_ok=False status=nan answer=nan ---
<think>
Okay, so I need to find the value of  $\frac{1}{r_1} + \frac{1}{r_2}$ 

```

5. Push to remote — status

Local commit cf8381c on branch stage8-probe-compare-results is ready. Push to github.com/Antony-zdh/Governor is blocked: the PAT in `/workspace/Governor/.env` authenticates as Antony-zdh (owner) but is a fine-grained token created without Contents-write permission on the Governor repo, so git push returns 403 (API: "Resource not accessible by personal access token"). A PAT with Contents: Read+write (or a classic PAT with repo scope) is required to complete the push.